

Limiting factors for lodgepole pine (*Pinus contorta*) and white spruce (*Picea glauca*) growth differ in some reconstructed sites in the Athabasca oil sands region



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ABSTRACT

Peat-mineral soil mix (PMM) over tailings sand (TS) or overburden (OB) is a commonly used prescription for oil sands reclamation. The TS and OB (the substrates) often have unfavorable conditions such as high salinity, soil compaction, low nutrient availability in OB, and low nutrient and water availabilities in TS, which may limit tree growth. The objective of this study was to identify limiting factors for growth of lodgepole pine (*Pinus contorta*) on PMM over TS and white spruce (*Picea glauca*) on PMM over OB in the Athabasca oil sands region, Alberta, Canada. In the pine sites, mean annual growth of height (HG), diameter at breast height (DBHG) and aboveground biomass (ABG) were significantly correlated with volumetric water content (VWC) in PMM and foliar $\delta^{13}\text{C}$ in current-year needles. Sites with high productivities had the lowest foliar $\delta^{13}\text{C}$ relative to sites with low and medium productivities. The VWC and dissolved organic carbon (DOC) together explained 80% of the variation in both the current annual increment of height (HI) ($p = 0.007$) and aboveground biomass (ABI) ($p = 0.008$). In the spruce sites, foliar potassium (K) and foliar and soil nitrogen (N) concentrations were positively correlated with HG, DBHG and ABG, consistent with patterns in K and inorganic N (the sum of $\text{NH}_4^{+}\text{-N}$ and $\text{NO}_3^{-}\text{-N}$) concentrations in PMM in those sites. Soil electrical conductivity (EC) and bulk density (Db) were greater in low than in medium and high productivity sites. The Db and inorganic N concentration together explained 67% of the variation in HI ($p = 0.037$), while Db and EC together explained 79% of the variation in ABI ($p = 0.009$). In conclusion, lodgepole pine trees planted on PMM over TS sites were limited by low water availability, while low soil N availability and high salinity limited the growth of white spruce trees planted on PMM over OB sites. Reclamation practices need to address water and nutrient limitations to ensure the success of reclaiming disturbed areas in the oil sands region.

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1. Introduction

Oil sand deposits in northern Alberta, Canada, contain about 168 billion barrels of bitumen, making it the third largest crude oil reserve in the world (Alberta Energy, 2014). The Athabasca oil sands region (AOSR) is the largest oil sands deposit in the province, in which crude oil can be extracted by open-pit mining (Fung and Macyk, 2000). Large scale open-pit mining in Alberta has disturbed approximately 767 km² of land, which represents 0.2% of Alberta's boreal forest (Government of Alberta, 2014a). The legislation requires that the disturbed land should be returned to equivalent

capability similar to that existed pre-disturbance (Government of Alberta, 2014b). The success of reclamation to upland forest after open-pit mining in the AOSR largely depends on the construction of landform and the placement of reconstructed soils. Reclaimed soils are reconstructed using peat-mineral soil mix (PMM) or LFH-mineral soil mix as capping soils above substrates such as overburden (OB) or tailings sand (TS) that are byproducts of the surface mining and oil sands extraction processes (Rowland et al., 2009; Naeth et al., 2013). However, reconstructed soils may lack growth conditions required for sustainable development of the reestablished vegetation.

Reconstruction of multi-layered soils can markedly change soil water availability for plant growth (Li et al., 2014). For example, the interface between the reclamation capping soil and the substrate restricts water movement between layers because of the low capillary action of TS or low permeability of OB that can be

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highly-compacted (Naeth et al., 2011; Jun et al., 2014). The dynamics of soil water content can greatly change during the growing season, making periodic soil water content measurement restricted for assessing soil water availability. However, foliar ^{13}C (carbon) isotope abundance ($\delta^{13}\text{C}$) has been shown to be a good indicator of long-term soil water availability, as foliar $\delta^{13}\text{C}$ integrates soil water availability over time and is closely related to plant water use efficiency (Sun et al., 1996). The closure of stomata under water limitation increases the fixation of $^{13}\text{CO}_2$ and reduces the discrimination against ^{13}C during photosynthesis, resulting in less negative $\delta^{13}\text{C}$ in plant tissues (Choi et al., 2005; Matsushima and Chang, 2007). The same tree species under different moisture regimes may have different $\delta^{13}\text{C}$ patterns (Sun et al., 1996; Zhang et al., 1997). We therefore employed the ^{13}C technique to study if water stress was limiting tree growth in the reconstructed landscape.

Nutrient availability, especially nitrogen (N) availability, can be another major factor that can restrict the reestablishment of plant communities. For example, planted trees growing on reconstructed mine sites were limited by low N availability in the substrates (Farnden et al., 2013). Mining and related activities may lead to severe loss of soil organic C and N because of the loss of topsoil and mechanical mixing of the A with B and C horizon soils (Ussiri and Lal, 2005; Shrestha and Lal, 2011). Although capping soils such as PMM and LFH-mineral soil mix have high organic N concentrations, the slow-release of organic N due to small microbial population sizes and reduced microbial activities may affect N availability (McMillan et al., 2007; Hemstock et al., 2010) and limit plant growth in the boreal forest region (Reeder and Sabey, 1987; Farnden et al., 2013). Low plant available N often limits the reestablishment and maintenance of vegetation on disturbed lands (Reeder and Sabey, 1987). Farnden et al. (2013) reported a positive relationship between tree height and soil organic matter content in a 21-year old jack pine stand in a reclaimed oil sands area, and they attributed the relationship to the contribution of organic matter to increased soil N availability. Although emissions of nitrogen oxides resulting from oil sands development in the last 40 years increased in the area surrounding oil sands mining, it was estimated that total N deposition into the ecosystems only ranged from 2.45 to 3.80 kg N ha $^{-1}$ yr $^{-1}$ in the AOSR (Laxton et al., 2010), which was lower than those in other parts of Canada and the United States (Watmough et al., 2005; Pardo et al., 2006), and many parts of Europe (Dise and Wright, 1995). In addition, N concentration and net N mineralization rates in soils were not affected by increased N emission (Laxton et al., 2010). Therefore, N deposition may not offer substantial relief for the demand for nutrients by the reestablished vegetation in oil sands reclamation.

Reconstructed soils in the AOSR are often affected by high salinity as well (Fung and Macyk, 2000). The salts, mainly composed of sodium, sulfate and chloride ions, normally migrate from the underlying processed TS or marine shale OB (Barbour et al., 2007). An upward movement of salts from the OB into the cover soil could raise the electrical conductivity (EC) level up to 6.0 dS m $^{-1}$ in the lower part of the cover soil (Kessler et al., 2010). Boreal tree species have considerable variability in salt tolerance between and within species (Allen et al., 1994; Khasa et al., 2002). High salinity can affect water availability and nutrient uptake, and thus may have detrimental effects on plant survival and growth and site productivity (McFee et al., 1981; Allen et al., 1994; Andrews et al., 1998; Rodrigue and Burger, 2004).

Reconstructed soils often possess other properties such as increased soil pH (Howat, 2000), heavy soil compaction, and high concentrations of residual bitumen (Fung and Macyk, 2000); those properties also can negatively affect the growth of trees. Poor tree growth can be identified by foliar discoloration, stunted growth, or other symptoms. The objective of this study was to identify the limiting factors for growth of two tree species planted for oil sands reclamation, lodgepole pine (*Pinus contorta*, Pl) planted in PMM over TS and white spruce (*Picea glauca*, Sw) planted in PMM over OB on reconstructed sites in the AOSR, Alberta. We hypothesized that the growth of Pl would be likely limited by water availability due to the coarse-textured TS substrate, while nutrient availability and salinity would be major limiting factors for the growth of Sw because of the saline OB substrate with low nutrient availability.

2. Materials and methods

2.1. Site description

The study area, located within the Suncor Energy Inc. Lease 86/17 (56°59'N and 111°32'W), is 22 km north of Fort McMurray in northeastern Alberta, Canada. The site is characterized by a continental boreal climate with short and cool summers and long and cold winters. The mean annual temperature is 0.7 °C, with mean daily temperatures ranging from −18.8 °C in January to 16.8 °C in July; the mean annual precipitation is 455.7 mm, with an average of 342.2 mm as rainfall during the growing season (Environment Canada, 2013). In the study area, 12 sites were set up in June 2011 (six Pl sites, site no. 1–6; six Sw sites, site no. 10–15) and 6 sites were set up in April 2012 (three Pl sites, site no. 7–9; three Sw sites, site no. 16–18), for a total of 18 sites, with 9 Pl sites on PMM over TS and 9 Sw sites on PMM over OB (Fig. 1). On each site, a 10 × 10 m plot was set up. The thickness of PMM and tree age varied from site to site (Table 1). The 9 sites for each tree species

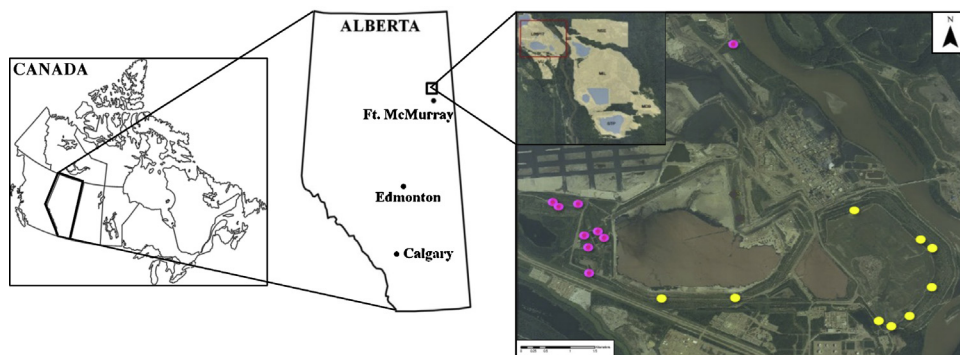


Fig. 1. Distribution of the studied lodgepole pine (*Pinus contorta*, Pl) and white spruce (*Picea glauca*, Sw) sites in Suncor Energy Inc. Lease 86/17 in the Athabasca oil sands region, Alberta, Canada. The yellow dots indicate Pl sites, and the purple dots indicate Sw sites. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

made up a site productivity gradient from low to high based on visual inspection of tree growth performance and the gradient was later confirmed by tree growth measurements. Understory plant communities on PI sites were dominated by prickly rose (*Rosa acicularis*), raspberry (*Rubus idaeus*), sweet clover (*Melilotus* sp.), dandelion (*Taraxacum officinale*), and slender wheat grass (*Agropyron trachycaulum*), while willow (*Salix* sp.), green alder (*Alnus crispa*), sweet clover, dandelion, and bluejoint grass (*Calamagrostis canadensis*) were the dominant understory species on Sw sites.

2.2. Tree growth and understory plant community measurements

Trees within each plot were tagged. Tree size was measured in September 2011 and 2012 for the first 12 established sites, and in May and September 2012 for the additional 6 established sites, assuming there was minimal growth during the winter season. Tree height (H) and diameter at breast height (DBH, 1.3 m above the ground) were measured using a height pole and a diameter tape, respectively. Aboveground biomass (AB) of trees on each site was calculated with DBH and height-based allometric equations described in Lambert et al. (2005) and Ung et al. (2008). Mean annual growth of height (HG), diameter at breast height (DBHG), and aboveground biomass (ABG) were calculated using the H, DBH and AB measured in 2012 divided by tree age. The differences in H, DBH, and AB of trees between 2011 and 2012 for sites established in June 2011, between May and September 2012 for sites established in April 2012, were calculated as the current annual increment of height (HI), diameter at breast height (DBHI), and aboveground biomass (ABI).

Five 1 × 1 m quadrats on each site were randomly established in August 2012 along the perimeter of the site to avoid sampling and other disturbance to understory plant community. Understory plant species including shrubs, herbs, grasses and mosses/lichens were identified, and the total cover for each category was visually estimated by the same person to minimize variation among operators. Total cover for understory community in each quadrat

was calculated by summing total cover for shrubs, herbs, grasses and mosses/lichens.

2.3. Soil and plant sampling

Soil samples were collected by layer in each site in July 2012. The top 20 cm PMM and the upper 20 cm substrate were collected using an auger in five randomly selected locations within each plot to obtain a composite PMM or substrate sample. If the thickness of the PMM was less than 20 cm, the entire PMM layer was collected. All soil samples were placed in plastic bags immediately after sampling, stored in a cooler and taken back to the laboratory. Bulk density (Db) of both PMM and the substrate (OB or TS) was determined by sampling soils from PMM or substrate layers in two soil pits in each site with a 100 cm³ stainless steel ring sampler. In October 2012, five healthy trees were randomly selected in each plot, and one branch from the upper one third of the live crown of each tree was collected to form a composite branch sample. Branch samples were placed in paper bags and taken back to the laboratory.

2.4. Soil and plant analyses

Upon returning to the laboratory, fresh soil samples were passed through a 2 mm sieve and manually homogenized after removing coarse fragments and roots. The samples were then analyzed for water content, NH₄⁺-N, NO₃⁻-N, and dissolved organic carbon (DOC) and dissolved organic nitrogen (DON). A subsample of each soil sample was air-dried for other chemical and physical analyses as described below.

The gravimetric water content of each soil sample was measured after oven-drying a subsample at 105 °C for 24 h and the volumetric water content (VWC) was calculated from gravimetric water content and bulk density. Soil texture was determined using the hydrometer method (Gee and Or, 2002). Soil pH was measured in deionized water using a digital DMP-2 mV/pH

Table 1

Characteristics of lodgepole pine (*Pinus contorta*, PI) and white spruce (*Picea glauca*, Sw) sites selected for this study in the Athabasca oil sands region, Alberta.

Site type ^a	Site no.	Year ^b	Density ^c (stem ha ⁻¹)	Peat-mineral soil mix					Substrate		
				Thickness (cm)	Bulk density (Mg m ⁻³)	Total C (g kg ⁻¹)	Total N (g kg ⁻¹)	Texture ^d	Bulk density (Mg m ⁻³)	Total C (g kg ⁻¹)	Texture
PI (TS)	1	1996	1900	17	0.77	67.1	2.7	SCL	1.50	0.9	S
	2	1991	2300	14	1.09	64.9	2.1	SCL	1.11	1.7	S
	3	1996	2300	14	1.04	50.2	1.3	SL	1.37	2.5	S
	4	1992	2700	30	0.84	78.8	2.9	SCL	1.47	2.2	S
	5	1991	2300	18	0.92	61.7	3.9	SCL	1.43	2.6	S
	6	1991	2100	24	0.64	159.6	3.7	SCL	1.58	2.9	S
	7	1992	2200	12	1.21	15.1	0.6	SCL	1.22	2.0	S
	8	1991	2100	22	1.42	84.9	3.6	SCL	1.61	1.6	S
	9	1996	2600	30	1.03	51.1	1.7	SCL	1.28	0.6	S
Sw (OB)	10	1982	1900	20	0.90	49.2	2.4	SCL	1.57	29.0	SCL
	11	1991	2300	11	1.24	44.9	1.7	SCL	1.48	42.4	SCL
	12	1991	3100	27	1.18	48.1	1.6	SCL	1.38	45.1	SCL
	13	1992	2800	16	0.76	73.5	2.6	SCL	1.06	38.9	SCL
	14	1992	1900	14	0.92	59.3	3.5	SCL	1.35	44.8	SCL
	15	1984	2500	25	0.32	235.1	10.5	N/A ^e	1.36	34.0	CL
	16	1991	2600	48	0.94	56.0	1.8	SCL	1.28	36.4	SCL
	17	1992	2300	26	1.03	170.5	9.3	SCL	1.46	43.3	SCL
	18	1992	2100	20	1.21	53.1	1.5	SCL	1.63	41.0	SCL

^a TS stands for a site reconstructed using tailings sand as a substrate below the peat-mineral soil mix and OB stands for a site reconstructed using overburden as a substrate below the peat-mineral soil mix.

^b Year is when PI and Sw trees were planted after soil reconstruction.

^c Density at the time of the study.

^d S, SL, SCL and CL stand for sand, sandy loam, sandy clay loam and clay, respectively.

^e N/A stands for not applicable because of high organic matter content in this site.

meter (Thermo Fisher Scientific Inc., Waltham, MA) with a soil to water ratio of 1:2 (w:v). Soil EC was measured using an AP75 portable waterproof conductivity/TDS meter (Thermo Fisher Scientific Inc., Waltham, MA) using 30 g of air-dried soil in 60 mL deionized water after shaking for 1 h at 25 °C and filtration through Whatman No. 42 filter papers. Concentrations of soluble cations including K^+ , Ca^{2+} , Mg^{2+} , Na^+ , and Al^{3+} were analyzed using a PerkinElmer Optima 3000 DV inductively coupled plasma mass spectrometer (ICP-MS) (PerkinElmer Inc., Shelton, CT) after extraction with deionized water at 1:2 of soil to water ratio (w:v) and filtration through Whatman No. 42 filter papers; a 1:4 ratio was used for the PMM from site 15 due to the high organic matter content in this site (Table 1).

Fresh soil samples were extracted with 0.5 mol L^{-1} KCl at 1:3 soil to solution ratio (w:v). After being shaken at 250 rpm on a mechanical shaker for 1 h, samples were filtered through Whatman No. 42 filter papers. The KCl extracts were steam-distilled with MgO to determine NH_4^+ -N concentrations using a Kjeldahl distillation system (Vapodest 20, C. Gerhardt, Königswinter, Germany). The extracts were distilled again after addition of Devarda's alloy to determine NO_3^- -N concentrations. A portion of each KCl extract was used for analysis of dissolved C and N concentrations with a TOC-Vcsn analyzer with TNM-1 (Shimadzu Corp., Kyoto, Japan). The total dissolved C concentration was the same as DOC concentration as no inorganic C was found in the soil. The DON concentration was calculated by subtracting NH_4^+ -N and NO_3^- -N concentrations from the total dissolved N concentration. Each air-dried soil sample was ground with a ball mill and used to analyze total C and N concentrations with a Carlo Erba NA 1500 elemental analyzer (Carlo Erba Instruments, Milano, Italy) at the Lethbridge Research Centre of Agriculture and Agri-Food Canada.

The foliage from each branch was separated into current-year and 1-year-old needles. After being quickly washed twice using distilled water, needles were oven-dried at 65 °C for 24 h or until constant weight. Three hundred needles were randomly selected and weighed. Needles were ground to pass through a 0.15 mm sieve in preparation for nutrient analyses. Concentrations of total C and N and ^{13}C in current-year and 1-year-old needles were determined with a stable isotope ratio mass spectrometer (IRMS) (Optima-EA, Micromass Ltd., Manchester, UK) linked to a Carlo Erba NA 1500 elemental analyzer (Carlo Erba Instruments, Milano, Italy). The ^{13}C isotope abundance was expressed as a relative deviation from the international standard of Pee Dee Belemite (PDB). Other total macro- (P, K, Ca and Mg) and micronutrients (Fe, Mn, Cu, Zn, B and Mo) in current-year needles were analyzed with the ICP-MS described above after digestion using concentrated HNO_3 and 30% H_2O_2 on a digestion block at 125 °C for 4 h (Campbell and Plank, 1998).

2.5. Statistical analyses

Correlation analyses were performed for HG, DBHG, ABG and foliar nutrients and soil properties, including soil VWC, pH, EC, Db, NH_4^+ -N, NO_3^- -N, DOC, DON, and PMM thickness. Nine sites for each tree species were classified into low, medium, and high productivity sites based on visual inspection of tree growth performance before tree size measurements. The productivity classification was later confirmed based on HG and DBHG. A one-way analysis of variance (ANOVA) was conducted to examine the statistical significance of a variable that was significantly correlated with tree growth along the productivity gradient. A regression analysis with stepwise selection for variables to be included in the model was conducted to determine the relationship between HI, DBHI, ABI and all soil variables. The significance level of both entering a variable into and removing a variable from

the model was set at 0.05. The stepwise regression equation for the HI, DBHI, and ABI of PI on PMM over TS or Sw on PMM over OB was set up as follows:

$$y = \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_j x_j + \beta_0$$

where y is HI, DBHI or ABI for PI or Sw; x_j is one of the soil properties; β_j is the coefficient of an estimate. An α value of 0.05 was chosen to indicate statistical significance in all analyses. All statistical analyses were conducted using the SAS software (SAS 9.2, SAS Institute Inc., Cary, NC).

3. Results

3.1. Lodgepole pine and white spruce growth and understory plant community cover along a productivity gradient

The DBHG ranged from 0.27 to 0.45 and 0.15 to 0.41 cm yr^{-1} on PI and Sw sites, respectively. The ABG ranged from 0.70 to 1.88 and 0.58 to 1.76 $\text{Mg ha}^{-1} \text{yr}^{-1}$ on PI and Sw sites, respectively (data not shown). The ABI of high productivity sites were 5.18 and 4.00 $\text{Mg ha}^{-1} \text{yr}^{-1}$ for PI and Sw, respectively, both of which were two times greater than those on low and medium productivity sites (Fig. 2A). Total understory plant community covers were greater in low and medium than in high productivity sites for PI ($p=0.042$), but not different for Sw (Fig. 2B).

3.2. Relationships between PI growth and VWC and foliar $\delta^{13}C$

The VWC were greater in high and medium than in low productivity sites (Fig. 2C). On PI sites, HG, DBHG and ABG were all positively correlated with VWC ($p=0.042$, 0.045 and 0.033, respectively) (Table 2). Foliar $\delta^{13}C$ in current-year and 1-year-old needles ranged from -28.2 to -25.4 and -27.9 to -25.2‰ , respectively, along the site productivity gradient. The PI sites with the highest productivity had the lowest foliar $\delta^{13}C$ in current-year needles, followed by those in medium and low productivity sites ($p=0.027$) (Fig. 2D), while there was no significant difference in foliar $\delta^{13}C$ of 1-year-old needles among the productivity levels ($p=0.640$). The HG and ABG were negatively correlated with foliar $\delta^{13}C$ in current-year needles ($p=0.036$ and 0.044, respectively) (Table 2). Based on the regression analysis, VWC and DOC together explained 80% of the variation in both HI and ABI ($p=0.007$ and 0.008, respectively) (Table 3), while there was no relationship between DBHI and any soil variables.

3.3. Relationships between Sw growth and foliar and soil properties

In Sw sites, HG, DBHG and ABG were all positively correlated with foliar N, foliar K, and soil inorganic N in PMM, and negatively correlated with Db. Only ABG was negatively correlated with soil pH in PMM ($p=0.047$), and positively correlated with soil NH_4^+ -N in PMM ($p=0.042$) (Table 2). The Db and soil inorganic N in PMM explained 67% of the variation in HI ($p=0.037$); the pH, EC and inorganic N in PMM explained 86% of the variation in DBHI ($p=0.013$), and the EC and Db in PMM explained 79% of the variation in ABI ($p=0.009$) (Table 3).

Foliar N and K concentrations in current-year needles in high productivity sites were greater than those in low and medium productivity sites ($p=0.038$ and <0.001 , respectively) (Fig. 3A and B). Foliar N concentrations in current-year needles in all sites were less than the critical value of 13 mg g^{-1} that indicates N deficiency (Ballard and Carter, 1986) except for the high productivity sites, while foliar K concentrations in current-year needles in all sites were greater than the critical value of 3.0 mg g^{-1} for K deficiency (Ballard and Carter, 1986).

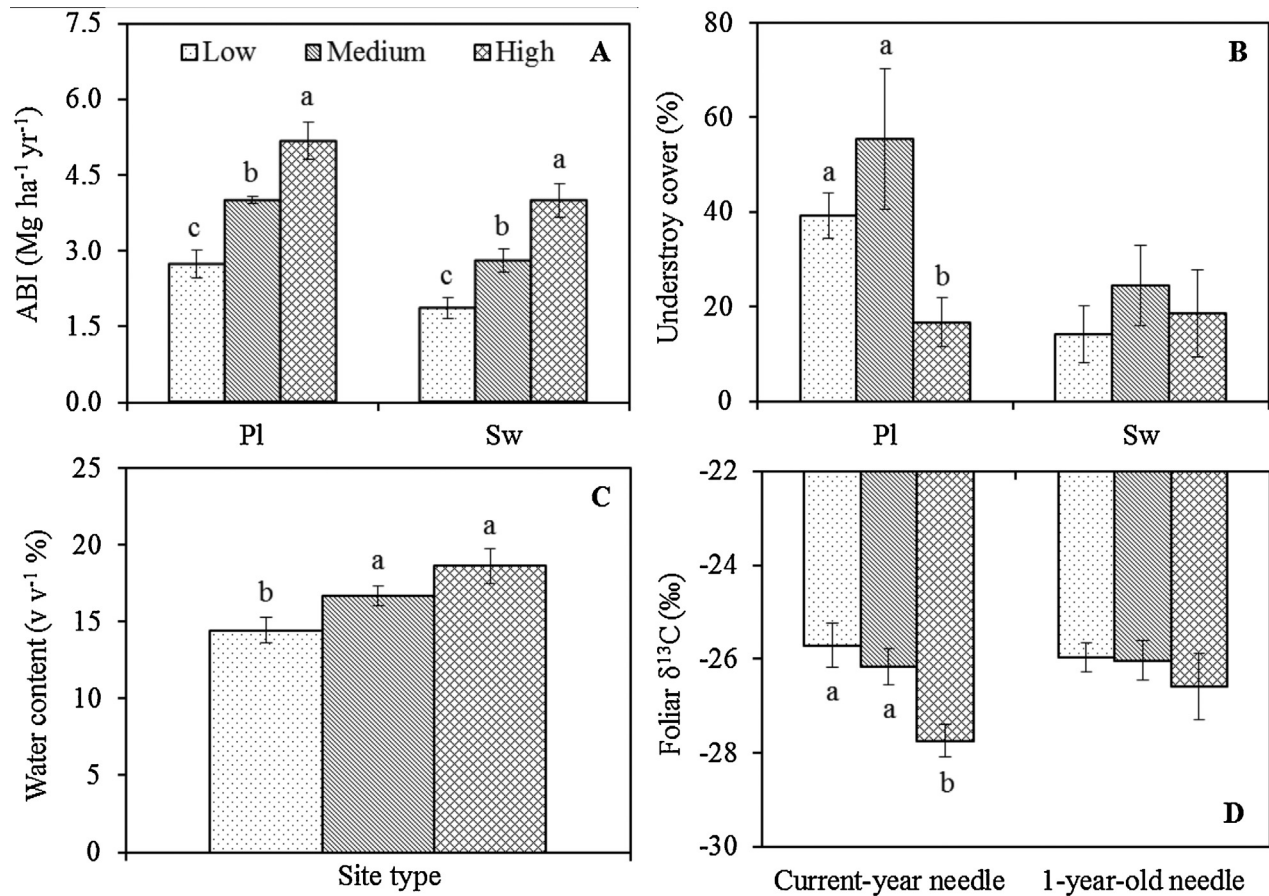


Fig. 2. (A) Increment of aboveground biomass (ABI) from 2011 to 2012 in lodgepole pine (*Pinus contorta*, Pl) and white spruce (*Picea glauca*, Sw) sites, (B) total understory plant community cover in Pl and Sw sites, (C) volumetric water content of peat-mineral soil mix (PMM) in Pl sites and (D) foliar $\delta^{13}\text{C}$ of current-year (2012) and 1-year-old needles of Pl. Different letters represent significant differences among different productivity levels at $p < 0.05$. Vertical bars are standard errors of means ($n = 3$).

High and medium productivity sites had greater soil $\text{NH}_4^+\text{-N}$, inorganic N and DON concentrations in PMM than low productivity sites ($p = 0.001$, 0.001 and 0.046 , respectively) (Fig. 4A), but $\text{NO}_3^-\text{-N}$ concentrations in PMM were not significantly different among the three productivity levels. There was no difference between high and medium productivity sites in $\text{NH}_4^+\text{-N}$, inorganic N and DON concentrations in PMM (Fig. 4A). The K concentrations in PMM were greater in high and medium than in low productivity sites

($p = 0.047$) (Fig. 4B), while not different between high and medium productivity sites.

There were significant differences in soil EC and Db among the productivity levels. High productivity sites had the lowest EC in PMM and OB, followed by medium and low productivity sites (Fig. 5A). The EC in PMM or in OB in low productivity sites was more than two times greater than that in high productivity sites, but EC in all sites were less than the salinity threshold of 4 dS m^{-1}

Table 2

Pearson correlation coefficients (r) and probability (p) for relationships between mean annual growth of height (HG), diameter at breast height (DBHG), aboveground biomass (ABG) and foliar nutrients and soil properties in lodgepole pine (*Pinus contorta*, Pl) and white spruce (*Picea glauca*, Sw) sites.

			Foliar $\delta^{13}\text{C}$	Foliar N	Foliar P	Foliar K	Foliar Ca	Foliar Mg	VWC ^a	Soil pH	Soil EC	Bulk density	PMM ^b thickness	NH ₄ ⁺ - N	NO ₃ ⁻ - N	Inorganic N	DON ^c	DOC ^d
Pl	HG	<i>r</i>	- 0.70^e	-0.10	0.22	0.32	0.27	0.36	0.80	0.03	-0.25	-0.38	0.09	0.26	0.38	0.35	0.35	0.09
		<i>p</i>	0.04	0.81	0.57	0.41	0.48	0.34	0.04	0.94	0.53	0.32	0.82	0.50	0.32	0.35	0.35	0.81
	DBHG	<i>r</i>	-0.59	-0.09	0.22	0.29	0.31	0.41	0.80	-0.04	-0.22	-0.44	0.02	0.41	0.56	0.53	0.49	<0.01
		<i>p</i>	0.10	0.82	0.58	0.45	0.41	0.27	0.05	0.92	0.56	0.24	0.96	0.28	0.11	0.14	0.18	1.00
	ABG	<i>r</i>	- 0.67	-0.11	0.31	0.18	0.19	0.23	0.93	0.03	-0.35	-0.43	-0.01	0.21	0.37	0.32	0.35	-0.05
		<i>p</i>	0.04	0.78	0.42	0.65	0.63	0.55	0.03	0.94	0.36	0.25	0.99	0.59	0.32	0.39	0.35	0.91
Sw	HG	<i>r</i>	-0.04	0.81	0.46	0.80	-0.16	-0.51	-0.20	-0.65	-0.61	- 0.78	-0.03	0.64	0.55	0.72	0.56	-0.28
		<i>p</i>	0.93	0.01	0.21	0.01	0.68	0.16	0.61	0.06	0.08	0.01	0.94	0.07	0.12	0.03	0.11	0.47
	DBHG	<i>r</i>	-0.08	0.77	0.40	0.72	-0.22	-0.49	-0.13	-0.51	-0.63	- 0.68	-0.06	0.63	0.58	0.73	0.48	-0.23
		<i>p</i>	0.85	0.02	0.28	0.03	0.56	0.18	0.74	0.10	0.07	0.05	0.89	0.07	0.11	0.03	0.19	0.56
	ABG	<i>r</i>	-0.05	0.85	0.54	0.84	-0.25	-0.49	-0.17	- 0.67	-0.65	- 0.79	-0.17	0.69	0.53	0.75	0.64	-0.23
		<i>p</i>	0.90	< 0.01	0.13	0.01	0.52	0.19	0.66	0.05	0.06	0.01	0.66	0.04	0.15	0.02	0.06	0.55

^a VWC, volumetric water content.

^b PMM, peat-mineral soil mix.

^c DON, dissolved organic nitrogen.

^d DOC, dissolved organic carbon. ^e Values in bold indicate significant correlation at $p < 0.05$.

^e Values in bold indicate significant correlation at $p < 0.05$.

Table 3

Regression equations developed based on the relationship between increment of height (HI), diameter at breast height (DBHI) and aboveground biomass (ABI) from 2011 to 2012 and soil variables through stepwise multiple regression analyses.

Site type	Growth parameter	Regression equation ^a
Pl	HI	HI = 0.025 VWC + 0.001 DOC – 0.278 ($R^2 = 0.80$, $p = 0.007$)
	ABI	ABI = 0.445 VWC + 0.014 DOC – 4.645 ($R^2 = 0.80$, $p = 0.008$)
Sw	HI	HI = –0.511 Db + 0.112 Inor.N + 0.610 ($R^2 = 0.67$, $p = 0.037$)
	DBHI	DBHI = –0.185 pH – 0.071 EC + 0.023 Inor.N + 1.750 ($R^2 = 0.86$, $p = 0.013$)
	ABI	ABI = –0.453 EC – 2.007 Db + 5.690 ($R^2 = 0.79$, $p = 0.009$)

^a VWC, volumetric water content; DOC, dissolved organic carbon; EC, electrical conductivity; Db, bulk density and Inor.N, inorganic N.

for tree growth in the AOSR except for the EC in OB in low productivity sites, which was 4.71 dS m^{-1} . High productivity sites had the lowest Db in PMM, followed by medium and low productivity sites (Fig. 5B). The Db of OB in low and medium productivity sites was 1.46 and 1.55 Mg m^{-3} , respectively; which was greater than in high productivity sites, but there was no significant difference between low and medium productivity sites.

4. Discussion

4.1. Limiting factors for lodgepole pine growth on PMM over TS sites

The significant positive correlation between VWC in PMM and Pl growth on PMM over TS sites suggested that water limitation was a factor affecting tree growth on those sites (Table 2). The total understory plant community cover was greater in low than in high productivity sites (Fig. 2B), suggesting that soil water availability may have been reduced due to high evapotranspiration under high vegetation cover. The understory vegetation could be a strong competitor with trees for the water stored in the soil, which could significantly decrease water availability for tree growth (Nambiar and Sands, 1993). Root distribution may also play a critical role in accessing plant available water in soils (Jung et al., 2014). In layered soils with distinct textural differences, the majority of roots in lodgepole pine sites were distributed in the fine-textured cover soil with good moisture availability and fertility, and the penetration of roots was inhibited by the coarse-textured soil layer (Horton, 1958). In a study conducted on the same sites as this study, fine root biomass significantly decreased in the tailings sand substrate

compared to that in fine-textured peat-mineral soil mix (Jung et al., 2014). The main distribution of roots in the shallow peat-mineral soil mix layer could decrease the accessibility to available water in deep soil layers and induce water stress under dry conditions.

Soil water storage and transmission characteristics are influenced by soil structure and texture (Barnhisel and Hower, 1997). Coarse-textured TS is generally a poor plant growth medium because of its low water and nutrient holding capacities and undesirable soil chemical properties that may be influenced by tailings water (Fung and Macyk, 2000); therefore it must be capped with organic materials such as PMM to improve water holding capacity and nutrient availability and to minimize the risk of contaminants contained in the substrates from affecting the establishment of vegetation on those reclaimed sites (Rowland et al., 2009). Although there is no standard prescription for reclamation, and oil companies adopt different capping depths based on the substrates used (Rowland et al., 2009), previous studies showed that placement of a thick capping soil layer with finer materials over coarser-textured TS layers can enhance water holding capacity due to the nature of the peat material and the capillary barrier effect (Moskal, 1999; Barbour et al., 2007; Li et al., 2014). The thickness of the PMM layer ranged from 12 to 30 cm, and was not different among low, medium and high productivity levels in this study, indicating that the PMM thickness was not the most limiting factor for the growth of Pl trees. Soil water storage is also influenced to a large extent by soil organic carbon content in capping soils. Positive relationships between soil water storage and PMM thickness or soil organic carbon content in PMM layer suggested that a thick capping soil layer with high organic matter

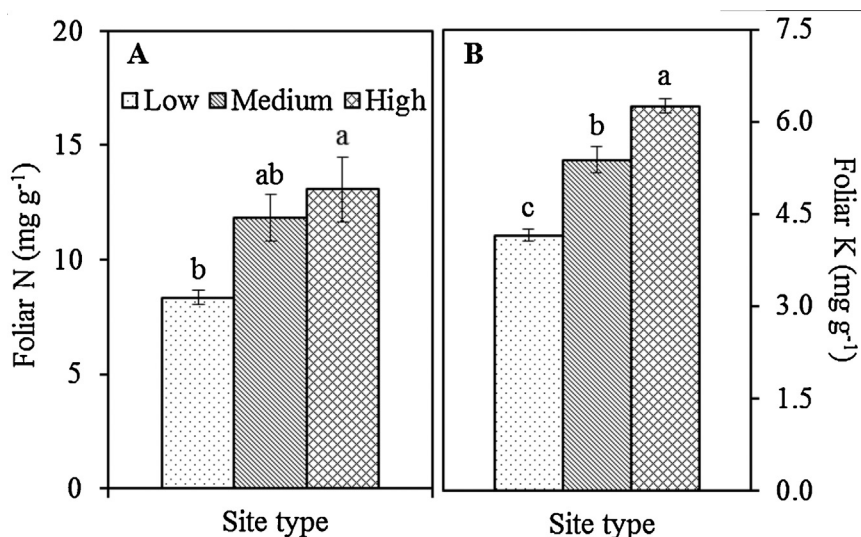


Fig. 3. (A) Foliar N and (B) K concentrations of current-year needles of white spruce (*Picea glauca*, Sw). Different letters represent significant differences among different productivity levels at $p < 0.05$. Vertical bars are standard errors of means ($n = 3$).

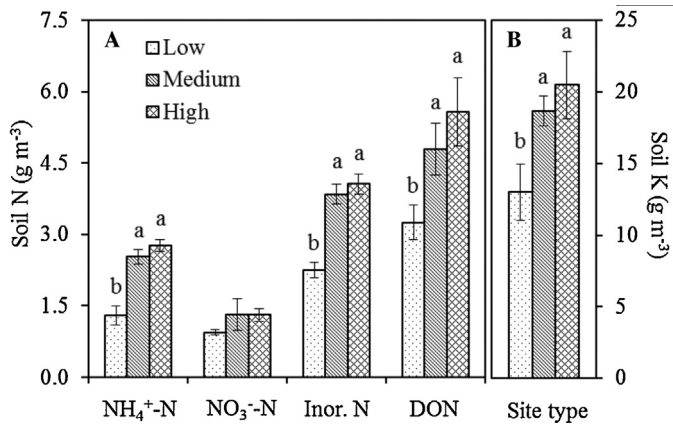


Fig. 4. (A) Soil N and (B) K concentrations of peat-mineral soil mix (PMM) in white spruce (*Picea glauca*, Sw) sites. Different letters represent significant differences among different productivity levels at $p < 0.05$. Vertical bars are standard errors of means ($n = 3$).

content could increase plant available water and reduce water stress in reclaimed soils (House et al., in preparation).

The significant difference in foliar $\delta^{13}\text{C}$ among productivity levels and significant correlation between foliar $\delta^{13}\text{C}$ and tree growth parameters indicated water limitation in PI sites (Table 2 and Fig. 2D). The low soil water availability in TS would limit fine root growth in the layer (Jung et al., 2014), thus restricting the ability of roots to absorb water and nutrients from the soil. High precipitation could increase soil water availability, resulting in increased stomatal conductance of leaves during photosynthesis, thus the plant could fix more $^{12}\text{CO}_2$ instead of $^{13}\text{CO}_2$, resulting in

higher discrimination against ^{13}C and more negative foliar $\delta^{13}\text{C}$ (Matsushima and Chang, 2007; Peri et al., 2012). Bowling et al. (2002) examined the variation in C isotopic composition in plant leaves in six coniferous forests across a moisture gradient (precipitation ranging from 227 to 2760 mm) in western Oregon, USA, and found that sites with greater precipitation had more negative $\delta^{13}\text{C}$ than sites with less precipitation. However, in this study, although the mean monthly precipitation during the 2011 growing season was 20 mm lower than that during the 2012 growing season (House et al., in preparation), lack of significant differences in foliar $\delta^{13}\text{C}$ between current-year and 1-year-old needles reflected that the extra 20 mm precipitation didn't significantly improve soil water availability.

The lack of difference of foliar N and K concentrations in current-year needles among productivity levels suggested that nutrient availability was not a limiting factor for PI growth. Foliar N concentration in current-year needles ranged from 13.9 to 14.5 mg g^{-1} (data not shown), and foliar K concentration in current-year needles ranged from 4.4 to 4.5 mg g^{-1} (data not shown). Both of those concentrations were greater than the critical value of 12 mg g^{-1} for N and 4.0 mg g^{-1} for K that indicate nutrient deficiency (Ballard and Carter, 1986). The moderate inorganic N concentration in PMM could contribute to sufficient N availability in PI sites. In addition, the root systems of PI trees were shallow and favored nutrient uptake from nutrient-rich surface organic layers (Jung et al., 2014). However, nitrogen availability was reported as a limiting factor for jack pine growth on PMM over TS site in the oil sands region (Farnden et al., 2013). This can be explained by different preferences of $\text{NH}_4^+\text{-N}$ and $\text{NO}_3^-\text{-N}$ uptake by tree species and different dominant inorganic N forms in the soil (Lavoie et al., 1992; Hawkins et al., 2008).

4.2. Limiting factors for white spruce growth on PMM over OB sites

Positive relationships between Sw growth and foliar N and K and the higher soil inorganic N and K concentrations in the high productivity sites indicated that N and K availabilities in the soil were limiting Sw growth on PMM over OB sites (Table 2 and Fig. 4). Foliar N concentration for optimal Sw growth has been reported to be between 15 and 25 mg g^{-1} (Nienstaedt and Zasada, 1990). However, foliar N concentration in this study was only 13 mg g^{-1} even in the high productivity sites, indicating that the Sw sites were N deficient. Greater $\text{NH}_4^+\text{-N}$ concentrations in medium and high than in low productivity sites and lack of differences in $\text{NO}_3^-\text{-N}$ concentrations contributed to foliar N differences among the productivity classes, as previous study reported that Sw had preference of $\text{NH}_4^+\text{-N}$ over $\text{NO}_3^-\text{-N}$ (Kronzucker et al., 1997). The dominant form of available N was $\text{NH}_4^+\text{-N}$ in this study, as compared to $\text{NO}_3^-\text{-N}$ as the dominant form of available N in reconstructed soils (Hemstock et al., 2010). The high salinity in the PMM in this study might have reduced net N mineralization (McClung and Frankenberger, 1987; Pathak and Rao, 1998) and inhibited net nitrification (Darrach et al., 1987; Inubushi et al., 1999), decreasing N availability and $\text{NO}_3^-\text{-N}$ concentrations in the soil. In addition, dramatic changes in soil physical and chemical properties after reclamation would influence microbial activity, organic matter decomposition, and lower net N transformation rates in reconstructed soils as compared to those in natural boreal forest soils (McMillan et al., 2007; Hemstock et al., 2010).

The average EC was 2.12 dS m^{-1} in the PMM layer and 3.05 dS m^{-1} in the OB layer (Fig. 5A). The average sodium adsorption ratio was 7.1 and 12.3 in PMM and OB layer, respectively (data not shown). The increased salinity in PMM layer may be caused by salt migration through convective flow and diffusion from OB to PMM layers (Merrill et al., 1980, 1983; Kessler et al., 2010). Although the EC and sodium adsorption ratio in the PMM in

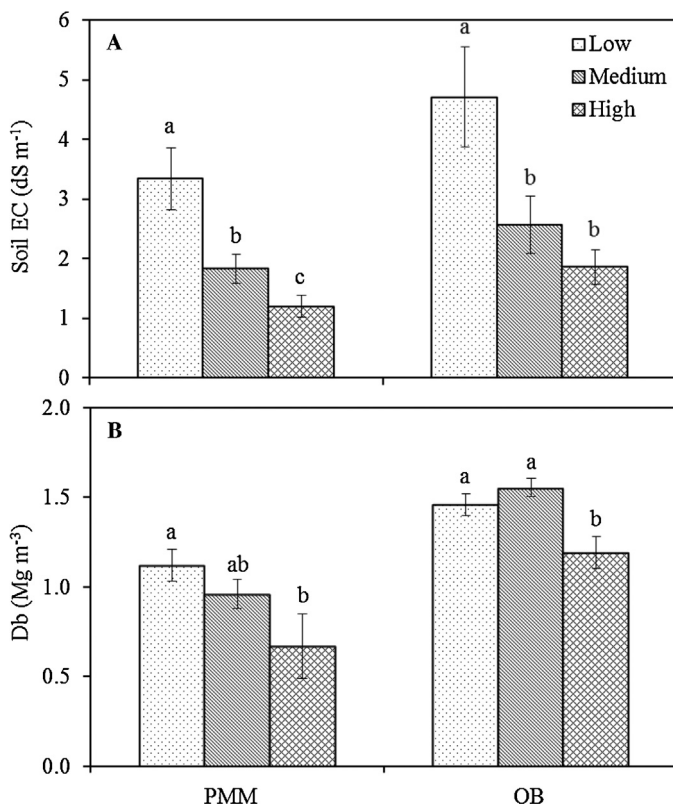


Fig. 5. (A) Soil electrical conductivity (EC) and (B) bulk density (Db) of peat-mineral soil mix (PMM) and overburden (OB) in white spruce (*Picea glauca*, Sw) sites. Different letters represent significant differences among different productivity levels at $p < 0.05$. Vertical bars are standard errors of means ($n = 3$).

all sites were less than the critical values of 4 dS m^{-1} and 15 for causing salt stress in tree growth in the AOSR (Howat, 2000), the roots would be affected by high salinity levels, particularly when available water was limited in the thin PMM layer. High salt concentrations, especially sodium, could reduce K uptake by plants (Renault et al., 1999), causing the lower foliar K concentrations on sites with higher salinity levels in this study. Munns and Tester (2008) reported that plants could increase the uptake of sodium and chloride ions in saline soils, which led to ion toxicity to plants and affected plant growth.

Soil compaction may also influence Sw growth. Previous studies showed that compaction decreased soil porosity, increased soil bulk density, reduced soil water holding capacity, restricted water infiltration, resisted root penetration to deep soil layers, changed nutrient cycling, and affected tree growth (Greacen and Sands, 1980; Corns, 1988; Cullen et al., 1991; Kozłowski, 1999; Startsev and McNabb, 2001; Tan et al., 2005). However, compaction of sub-layers could reduce water infiltration and increase water storage in soil layers above, increasing the amount of available water for plant uptake (Jung et al., 2014). This may explain the lack of a relationship between soil water content in PMM and Sw growth on OB sites, with the high bulk density in the OB layer and increased water availability in the PMM layer in this study. Although the Db in the TS layer was not significantly different from that in the OB layer, the high Db in TS was due to the sandy texture of TS, while the high Db in OB was from soil compaction by heavy equipment or from highly compacted reconstructed materials; the same Db in TS and OB had different pore size distributions, which would influence the expansion of root systems in different ways. The roots could penetrate deep into the TS layer but not the OB layer because of greater macro porosity in TS than in OB.

5. Conclusions

The growth of lodgepole pine on sites reconstructed using the PMM as a capping soil over the TS substrate was mainly affected by water availability, while N availability, salinity and soil compaction combined limited Sw growth on reconstructed sites using the PMM over the OB substrate. Effects of water and nutrient availabilities on tree growth and effects of salinity on N availability should be taken into consideration when planning reclamation prescriptions, which require appropriate capping depth over substrates to reduce the upward movement of salts to the rooting zone and at the same time increase water storage for tree growth. Fertilization may be used to increase nutrient availability and improve tree growth performance in some of the low productivity sites. Therefore, reclamation of disturbed oil sands sites should consider multiple factors that may limit the success of land reclamation to ensure the sustainable development of reconstructed oil sands sites.

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